2 **(a)** Fig. 2.1 is a graph that shows how the extension of a spring varies with the load suspended from it.

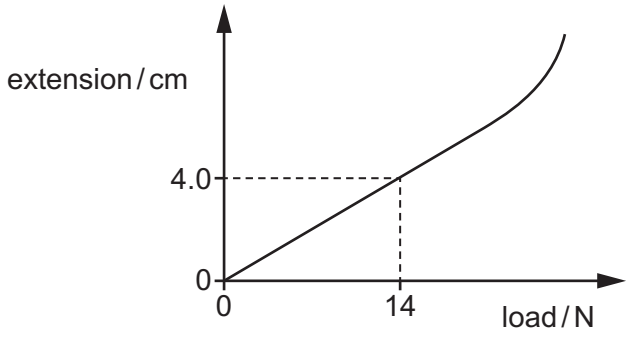


Fig. 2.1

(i) Determine the spring constant of this spring.

spring constant = ... [3]

(ii) On Fig. 2.1, mark the limit of proportionality and label this point L. [1]

(b) Fig. 2.2 shows a car travelling at constant speed around corner A on a road.

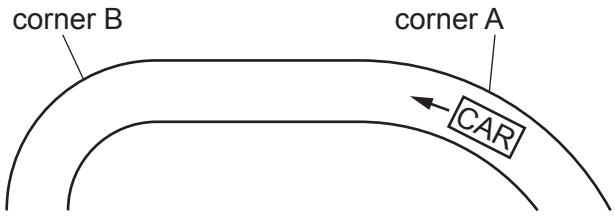


Fig. 2.2

(i) On Fig. 2.2, mark with an arrow the direction of the resultant force acting on the car as it travels around corner A. [2]

(ii) Corner B has a smaller radius than corner A. The car travels at the same speed around corner B as around corner A.

State how the resultant force changes due to the car travelling around a corner of smaller radius.

..... [1]

[Total: 7]

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